2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT) | 979-8-3503-7024-9/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICCCNT61001.2024.10724295

# Transformer-based Transfer Learning for Enhanced Speech Dysarthria Severity Assessment

Venkata Siva Manoj A*, Velaga Lakshman*, Akash Kamuju*, Venu Pulagam*, Jyothish Lal G*
*Amrita School of Artificial Intelligence, Coimbatore, Amrita Vishwa Vidyapeetham, India
{addalavenmanoj, velagalakshman15, akashkamuju02, notvenupulagam}@gmail.com,
g_jyothishlal@cb.amrita.edu

***Abstract*—Dysarthria, a neuromuscular communications disorder presented with impaired pronunciation, is a daunting task to identify and quantify the level of dysfunction. Through this paper, a critical study of automated dysarthria severity classification using transformer-based deep learning methods will be discussed. Transfer learning is facilitated by employing three variations of Vision Transformer (ViT) models: ViT-L-16, ViT-L-32, and ViT-B-16, on UA-Speech Corpus and TORGO datasets, achieving remarkable results with 99.01% accuracy for the UA-Speech dataset and 99.39% accuracy for the TORGO database. The study evaluates the models' performance focusing on accuracy, precision, recall, and F1-scores. Experimental results emphasise the potential for automated diagnostic units in neurology clinical practice and establish a baseline for leading work in dysarthria severity classification with the selection of an efficient ViT model.**

***Index Terms*—Transfer Learning, Speech Processing, Dysarthria Severity Detection, Transformers.**

## I. Introduction

Dysarthria, a motor speech disorder in which there is difficulty in articulation as a result of muscle weakness, paralysis, or inability to coordinate the muscles involved in speech production. It happens due to tissue damage of the nervous system that controls the speech muscles which affects the ability to pronounce speech sounds clearly. It can express itself in different forms, such as slurred, unclear speech, voice changes, and inability to control pitch and tone of voice. A number of neurological disorders like amyotropic lateral sclerosis, Parkinsons disease, Cerebral palsy, and trauma to the nervous system can all be caused by dysarthria [1], [2]. The articulation difficulties resulting from Dysarthria cause the reduction of vocal message comprehension degree through the reduced intelligibility, audibility, naturalness, and efficiency of oral communication. Dysarthria disrupts multiple components of the speech production process, resulting in poor phonation, prosody, and articulation [3]. Dysarthria can cause challenges with society, thoughts of isolation, despair, along with various psychological issues [4], [5]. As a result, if immediate assistance and early treatment are not provided, disease course management would be difficult, and the condition will progress further. Doctors may detect dysarthria primarily but it is often regarded as a costly, arduous, and time-consuming diagnostic [6].

The degree of dysarthria(severity level) should be evaluated as a critical diagnostic step, since this is of great importance for understanding the course of the underlying condition and guidance of clinical decisions regarding therapy or drugs when needed. Also the periodical speech therapy sessions are planned in accordance with it and then they are conducted. Speech-language pathologists are often trained to use intelligibility tests in conjunction with voice clinics to conduct assessments. Fortunately using such an approach can be expensive and variable [7]. The implementation of a standardised objective severity evaluation is useful in a variety of ways, including cost-effectiveness and consistency.

Speech assessment as a diagnostic technique holds considerable potential in neurology and oftentimes is linked with diagnosing complex neurological disorders like dysarthria. These disorders usually generate their own distinct speech patterns, which can be utilised as key diagnostic markers. Speech analysis is one of the most important simple and economic methods for early detection and management of various neuro-disorders [8]. Medical professionals may be able to aid with early diagnosis and treatment planning by carefully examining such speech traits, which may result in an increase in patient quality of life and the ability to provide timely treatments.

To the best of the authors' knowledge, this is the first study which utilizes a transfer learning approach to categorise dysarthria severity into very low, low, medium, and high levels, depending on the acoustic features and neural network-based learning techniques. This study presents transformer-based deep learning architectures for classifying dysarthria severity using speech indicators. The use of transformer-based models in speech dysarthria severity assessment is highly significant due to their advanced capabilities in feature extraction and handling long-range dependencies within speech data. These models, particularly Vision Transformers (ViTs), excel at capturing complex patterns and subtle variations in speech spectrograms, which are critical for accurately identifying different levels of dysarthria severity. Their robustness to data variability and the ability to leverage transfer learning reduce the dependency on large, task-specific datasets, enhancing performance even with limited data. Experimentally, transformer models have shown superior accuracy, precision, and recall, significantly outperforming other architectures. Moreover, transformers' scalability and flexibility allow for easy integration with multimodal data, potentially improving assessment accuracy further. Overall, the adoption of transformer-based models sets a new benchmark in dysarthria severity assessment, driving



rized licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:48:43 UTC from IEEE Xplore. Restrictions a

future research and innovation in clinical speech assessment practices. Then, a detailed comparison study is carried out to determine the most finely tuned deep learning architecture that handles the multi-class classification challenge.

The following sections of this paper are organised as follows: Section II contains a brief description of the literature review, Section III describes about the data used and provides a thorough overview of the methodology, Section IV shows the results, and Section V presents the conclusion and future scope.

## II. Literature Survey

Many automated dysarthria severity level classification methods rely on acoustic data to identify the key aspects of dysarthric speech production [9], [10]. The research in [11] emphasizes the importance of selecting the right deep learning architecture, which is illustrated by the fact that accuracies of 96.18% and 93.24% were achieved with different DNN, CNN, and LSTM models on the Universal Access Dysarthric Speech corpus and TORGO data. That is in contrary to the SVM classifiers and hence the model selection should be taken seriously in order to improve performance. In [12], by delving into integrity and validity of ten speech-language pathologist's, the speech severity of 52 amyotrophic lateral sclerosis and 20 healthy speakers was rated. Thresholds for dysarthria severity were determined by Receiver operating characteristic (ROC) curves. Difficulties arose when opting for a stratification indicator. To improve the unbiased estimation of intelligibility in speech, Mahalanobics distance based discriminant analysis classifier was developed to classify into two levels which was trained on 34 acoustic features, achieving 95% accuracy in [13]. Integrating the mappings formulated by selecting nine prominent features, results in enhancement up to 13% in accuracy. [1] uses ANN with Multitapered spectral estimation based features and audio descriptors as attributes for classifying speech severity.Examination of UA-Speech corpus and TORGO database brings about 96.44% and 98.7% accuracy.

Further in [14], utterances of 13 healthy speakers and 19 dysarthrics detected speakers from Universal Access Speech database were employed as mel spectrograms and given to a residual CNN. To recognize the salient periods in the speech, multi-head attention mechanism was incorporated. Uplift of 11.51% in accuracy was observed compared to the initial system. A cross-database approach is implemented in [15], where models were trained on UA-Speech and tested on Atlanta Motor Speech Disorders Corpus divisions. To record distinctive features of dysarthric speech Teager Energy Operator (TEO) and glottal waveform were embedded, which resulted in 90% accuracy. Digital signal processing algorithms are employed for evaluating severity of features in [16]. Linear discriminant classifier and non-linear classifier are utilized for classification. Non-linear classifier surpasses linear classifier which additionally delivers bi-dimensional representation of the data. In [17] discerning dysarthria and evaluating its severity was by inspecting the rhythm metrics. For the QoLT Korean dataset, comparing accuracies acquired from rhythm-based features utilising Random Forest, Support Vector Machine, and feed-forward neural network to those obtained from prosodic features found 7.5% and 15% gains in identification and severity assessment, respectively. In parallel with other studies, pronunciation is taken into account as an additional factor for extracting features in [18], thus characteristics of dysarthric speech are accurately depicted. Percentage of correct phonemes and degree of vowel distortion assessed as pronunciation features. In [19] Hierarchical cluster analysis is implemented on twenty-five-year-old children with dysarthria secondary to CP and twenty normally progressing children as a control group to predict the organic subgroup of children derived from six acoustic and perceptual speech measures to measure segmental and suprasegmental speech characteristics. [20] tried to elevate the efficacy and applicability of speech-powered tools by proposing a Residual Network-based strategy on standard Universal Access corpus, considering short duration speech segments as input features and comparison was made against CNN, GMM and Light CNN's. Dysarthria arisen from Amyotrophic Lateral Sclerosis was dealt in [21]. Data from 119 ALS and healthy speakers was employed to conduct automatic classification of vowels and fricatives at diverse degrees of severity of dysarthria and appraise the findings with manual classification, where boost in accuracy was found.

The review presents the state of the art in fully automated dysarthria severity analysis, showing the different approaches and techniques used in the researches journals. Studies have come to use the acoustic attributes, deep learning architectures, and speech assessment techniques to evaluate dysarthria severity levels correctly. Nevertheless, the ranking achieved is relatively low in terms of accuracy and scalability owing to multi-class classification and database heterogeneity Based on the foundation started by using transfer learning with deep-neural networks we intend to improve the severity detection of dysarthria. Our novel idea is based on the application of deep learning transformer architecture based on transfer learning. In addition, our investigation is a milestone for upcoming research in dysarthria categorization which would rely on open-source datasets and standard evaluation indicators.

## III. Materials and Methodology

In this section, we provide the materials and approach utilized in our work for diagnosising the severity of dysarthria through transfer learning with transformers. The datasets used for this study are including the UA-Speech Corpus and the TORGO dataset, along with an overview of the methodology adopted for model training and evaluation. Furthermore, the transformer based deep learning architectures selected for transfer learning are outlined, along with the simulation configuration and evaluation metrics used to assess model performance.

### A. Data

The proposed work was validated with the Universal Access Dysarthric Speech Corpus (UA-Speech), a standard American



rized licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:48:43 UTC from IEEE Xplore. Restrictions a

English dysarthric database [9] and TORGO dataset [22].

*1) UA Dysarthric Speech Dataset:* The UA-Speech dataset contains the recordings of 13 healthy individuals and 19 dysarthria sufferers, but only the data from 15 patients were accessible. Each participant contributed recordings of 155 common words, repeated three times, resulting in 465 common words per speaker. In addition, the corpus includes 300 different rare words per speaker, gathered through a greedy algorithm from children's novels digitized by Project Gutenberg and according to a sequence pooling criterion. Data from the sixth microphone channel, sampled at a frequency of 16 kHz, were utilized due to its superior signal-to-noise ratio. The intelligibility scores provided by five naive listeners for each speaker in UA-Speech were used to assign the degrees of severity of very low, low, medium, and high.

*2) TORGO:* The TORGO dataset contains healthy person's 3D articulator markers and speech records acquired simultaneously, along with articulatory and acoustic data from those persons suffering from cerebral palsy. It entails auditory stimuli of nonwords, short words, and phrases of known and unknown interchange, acquired using two types of microphones, an array one, and a wearable one. The data sets includes the twenty-six English digits, international radio alphabets, some common English words, and phonetically contrasting pairs to provide a wide range of speaking stimuli. The Perceptual dimensions of speech were assayed using the Frenchay Dysarthria Assessment (FDA) [23] that is a standardized assessment meant to assess various motor aspects of speech. With 16 kHz head-mounted microphone recordings, the computer-generated TORGO dataset has been collections for speech dysarthric speech studies towards understanding both acoustic and articulatory elements across different speech tasks and participant groups [22].

Table I summarises the severity categories for the UA-Speech and TORGO datasets.The UA-Speech dataset uses intelligibility to categorise severity, but in the TORGO database, it relies on the total FDA score for dysarthria patients, as referenced in [24]. Each voice recording is subsequently converted to a mel-spectrogram and scaled to 224x224 and input into Deep Learning transformer model as shown in the Figure 1. Figure 2 shows how the dataset is divided into training, testing, and validation.

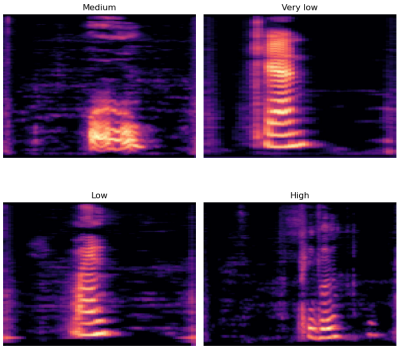



Fig. 1. Spectrograms Representing Different Dysarthria Severity Classes

TABLE I
DEGREE OF SEVERITY IN UA SPEECH AND TORGO DATABASES

| Degree of Severity | UA-Speech | TORGO |
|---|---|---|
| Very Low | F05, M08, M09, M10, M14 | F03, F04, M03 |
| Low | F04, M05, M11 | F01, M05 |
| Medium | F02, M07, M16 | M01, M02, M04 |
| High | F03, M04, M12, M01 | - |

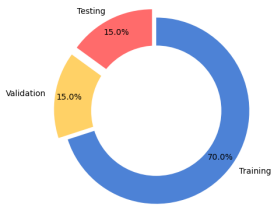



Fig. 2. Visualisation of the dataset partition for training, testing, and validation.

## B. Methodology

The effectiveness of transfer learning in meeting this challenge was investigated. To do this, three variants of the Vision Transformer deep learning model were used, each augmented with a classification head. The classification head has six fully connected layers with ReLU as the activation function. The whole model, including the classification head, was fine-tuned employing the Adam optimizer at a learning rate of 0.001. Training continued for all three types until they converged, with evidence of overfitting appearing after around 150 epochs. The variant of the Vision Transformer model with the best validation accuracy was then determined retrospectively. All test measures in the study are based on the variant that has the best validation accuracy. Figure 3 shows the proposed approach for the classification problem.

A classification head consisting of six linear layers with ReLU activations was added to each of the three Vision Transformer deep learning model variations that were used. The basic model and the classification module of the overall architecture were fine-tuned at a learning rate of 0.001 using the Adam optimizer.

The proposed methodology for dysarthria severity classification using transformer-based transfer learning can be readily adapted to assess various other speech and language disorders. By selecting disorder-specific datasets and preprocessing recordings to highlight relevant features, such as stuttering repetitions, aphasia linguistic errors, or apraxia speech motor planning deficits, the model's versatility can be leveraged. Fine-tuning the pre-trained transformer model on the new datasets, adjusting the architecture and hyperparameters as needed, ensures its suitability for the target disorder. Evaluation metrics tailored to the disorder's severity levels, coupled with rigorous validation and testing, validate the model's performance. This adaptable methodology offers a promising avenue for enhancing diagnostic accuracy and supporting clinical decision-making across a range of speech and language disorders.



rized licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:48:43 UTC from IEEE Xplore. Restrictions a

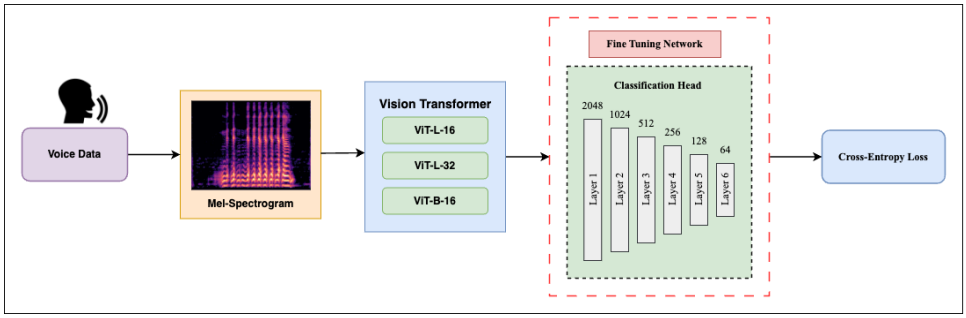



Fig. 3. Flow Diagram of the Proposed Approach

*1) Simulation Configuration:* The training was done with the help of the PyTorch library on the Kaggle platform using a T4 GPU acceleration for processing speed. The chosen hyperparameters were an ADAM optimizer with a learning rate of $10^{-3}$, 150 epochs, and a batch size of 32.

### C. Model

ViTs, short for Vision Transformers [25]–[27], are astounding in the way they boost image recognition by replacing convolutions with the self-attentive mechanism. The fact that ViTs understand images as essentially nothing but sequences of tokens allows them to discover the corresponding global context, including possible ties, which, in turn, facilitates their feature extraction. The scalability gives them flexibility, being able to customize the resolution and dividing the input images into blocks and linear bandwidths. The implemented novel architecture of this proposed vision system facilitates a leap forward in computer vision, providing more adaptability and efficiency for the analysis of images.

ViT-B-16, ViT-L-16, and ViT-L-32 are the names of the three variants that make up the innovation. ViT-B-16 has the same base configuration but is computationally less expensive. Therefore, it is a suitable model for systems that have limited computational resources to perform the task. Vit-L-16 makes use of a bigger model size to increase its capability to characterize in-depth characteristics and likewise the fine features of the images. As opposed to ViT-B-16, the ViT-L-32 model offers a larger size allowing for better representation space and performance achieving the largest possible image analysis tasks. There is a different variant available which contains that particular feature expected in a numerous application scenario. The choice of variant significantly contributes to the enhanced performance of vision transformers.

### D. Evaluation Metrics

This study utilizes the following assessment metrics: Accuracy, F1-measure, precision, and recall. The Accuracy gives a full picture of a model's prediction correctness, whereas the F1-score is a balanced measure of precision and recall, which is critical when working with unbalanced datasets. Precision expresses the accuracy of positive predictions regarding the true positives within all the predictions that are positive, whereas recall evaluates the fraction of the relevant instances correctly identified by a model compared to those that it missed among the total positive instances. The metrics of evaluation used in this work are presented in Table II.

TABLE II
ERROR METRICS UTILIZED TO EVALUATE THE PERFORMANCE OF THE DEEP LEARNING MODEL UNDER CONSIDERATION IN THIS INVESTIGATION.

| S.No | Error metric | Formula |
|---|---|---|
| 1 | Accuracy | $\frac{TP+TN}{TN+FP+TP+FN}$ |
| 2 | F1 score | $\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$ |
| 3 | Precision | $\frac{TP}{TP+FP}$ |
| 4 | Recall | $\frac{TP}{TP+FN}$ |

Where TP represents the count of true positives, FP denotes the count of false positives, TN signifies the count of true negatives, and FN indicates the count of false negatives.

## IV. Experimental Results

The study of the dysarthria severity spectrum, with a focus on the application of transfer learning on voice recordings using transformer models, shows significant improvements in performance metrics of these models. The results in the Table III and Table IV show that the ViT-L-32 model is the superior version as it provides the best accurate results in both the UA-Speech and TORGO datasets, respectively, as the model can achieve accuracies of up to 99.01% and 99.39%, respectively. It gets a place right behind the VI-B-16 with their scores only 98.66% and 98.78% for UA-Speech and TORGO datasets, respectively. ViT-L-16 is also among excellent models and its performance proves to be not less perfect, reaching 98.6% and 98.12% On the UA-Speech and TORGO datasets.Table V provides the class-wise F1 scores of the ViT-L-32 model, which shows in detail the model's performance in different severity levels.

The preeminence of the ViT-L-32 model in discerning dysarthria severity is further elucidated through the comprehensive analysis of confusion matrices of both TORGO and UA-Speech datasets, as depicted in Figures 4 and 5, respectively. The scores of these tables show a lot of true


rized licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:48:43 UTC from IEEE Xplore. Restrictions a

TABLE III
AVERAGE PERFORMANCE METRICS OF MODELS (UA-SPEECH DATASET, AVERAGED OVER 4 RUNS).

| Model | Accuracy | F1 | Precision | Recall |
|---|---|---|---|---|
| **ViT-L-32** | **0.9901** | **0.9900** | **0.9900** | **0.9899** |
| ViT-B-16 | 0.9860 | 0.9866 | 0.9854 | 0.9865 |
| ViT-L-16 | 0.9866 | 0.9867 | 0.9870 | 0.9864 |

TABLE IV
AVERAGE PERFORMANCE METRICS OF MODELS (TORGO DATASET, AVERAGED OVER 4 RUNS).

| Model | Accuracy | F1 | Precision | Recall |
|---|---|---|---|---|
| **ViT-L-32** | **0.9939** | **0.9927** | **0.9934** | **0.9922** |
| ViT-B-16 | 0.9878 | 0.9871 | 0.9861 | 0.9882 |
| ViT-L-16 | 0.9818 | 0.9802 | 0.9798 | 0.9806 |

positives and true negatives, thus indirectly illustrating that the model functions well as a classifier. Furthermore, the ViT-L-32's efficacy in capturing subtle patterns is vividly illustrated in the loss plots, showcased in Figure 6 for TORGO dataset and Figure 7 for UA-Speech dataset.

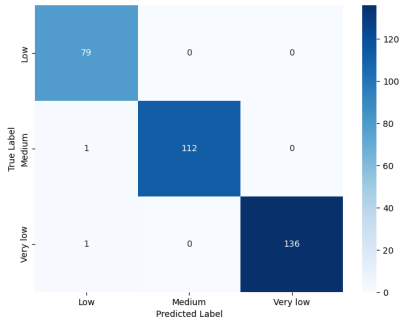



Fig. 4. Confusion matrix generated by ViT-L-32 for performance assessment on the TORGO dataset.

The distinctive performance of ViT-32 can be traced back to the transformer architecture which shows an advantage in processing data such as voice recordings, allowing the example of dysarthria severity classification to indicate that much more pattern recognition is required, which can only be achieved through learnable deep networks. On the other end, maybe it is the restricted dataset characteristics that are accounting for the comparative lack of progress of ViT-B-16 and a search for more optimization strategies is needed.

Albeit these, the results illuminate the fact that model choice linked with the characteristics of the dataset as well as the computational constraints is a key consideration. The use of such formative transformer models with ViT-L-32 is that it facilitates the identification of the minute acoustic cues by making it possible to extract the vocal recordings. Apart from that, they emphasize the idea that it is impossible to just use the fact that a new model is better than another to claim that it will definitely outperform in tasks where the older models outperforms the newer ones.

Additionally, the study has derived that speech disfluency examination by these models is based on their proficiency to recognize subtle variations in subharmonics within voice

TABLE V
CLASS-WISE F1 SCORES ACHIEVED BY THE VIT-L-32 MODEL ON THE UA-SPEECH AND TORGO DATASETS.

| Class | F1-Score | |
|---|---|---|
| | UA-Speech | TORGO |
| Very Low | 0.9884 | 0.9963 |
| Low | 0.9841 | 0.9875 |
| Medium | 0.9903 | 0.9956 |
| High | 0.9834 | - |

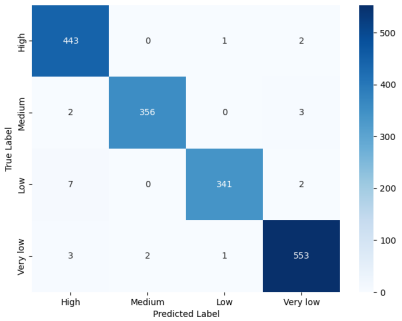



Fig. 5. Confusion matrix generated by ViT-L-32 for performance assessment on the UA-Speech dataset.

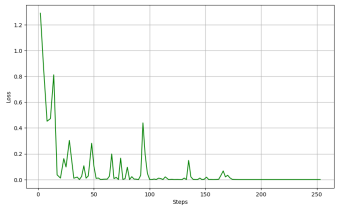



Fig. 6. Visualization of validation loss of ViT-L-32 on the TORGO dataset.

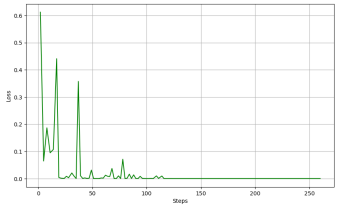



Fig. 7. Visualization of validation loss of ViT-L-32 on the UA-Speech dataset.

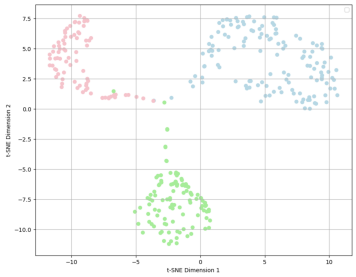



Fig. 8. The t-SNE plot depicting the test data representation by ViT-L-32 on the TORGO dataset.



rized licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:48:43 UTC from IEEE Xplore. Restrictions a

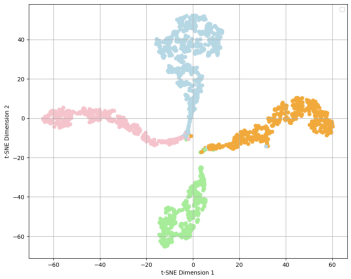



Fig. 9. The t-SNE plot depicting the test data representation by ViT-L-32 on the UA-Speech dataset.

recordings. This supposition is supported by the t-SNE plots as depicted in figures 8 and 9 for TORGO AND UA-Speech datasets, repectively. This reveals the ease with which the model differentiate and absorb various features that are characteristic for severity level in both UA-Speech and TORGO datasets.

## V. Conclusion

In the current study, we put forward an idea to use the fine-tuned ViT-L-32, a deep learning model with transformers as a base module, aiming to meet the goal of recognizing the levels of dysarthria. Our networks are fed with Mel-spectrograms, with the aim of creating a classifier with better metrics. From the pool we have assessed, the model ViT-L-32 stood out, obtaining an accuracy of 99.01% and 99.39% for the UA-Speech and TORGO datasets, respectively.

Further, there are many possible ways to expand the scope of this research as the work progresses. One of the avenues is to look for greater coverage of the dataset to include more varied patients to improve the model‘s accuracy and real world applicability. Beside that, the model deployment in real-time applications and its tracking of the performance across multiple datasets which includes different data distributions will address its generalization performance and practical utility.

## References

[1] C. Bhat, B. Vachhani, and S. K. Kopparapu, “Automatic assessment of dysarthria severity level using audio descriptors,” in *2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 5070–5074, IEEE, 2017.

[2] A. Mohamed Ali, G. Jyothish Lal, V. Sowmya, and E. Gopalakrishnan, “Parkinson’s disease assessment from speech data using recurrence plot,” in *International Conference on Computing, Intelligence and Data Analytics*, pp. 132–142, Springer, 2023.

[3] F. L. Darley, A. E. Aronson, and J. R. Brown, “Differential diagnostic patterns of dysarthria,” *Journal of speech and hearing research*, vol. 12, no. 2, pp. 246–269, 1969.

[4] S. Marmor, K. J. Horvath, K. O. Lim, and S. Misono, “Voice problems and depression among adults in the u nited s tates,” *The Laryngoscope*, vol. 126, no. 8, pp. 1859–1864, 2016.

[5] P. N. Chowdary, M. S. Akshay, V. S. Aravind, M. S. Aashish, G. V. V. Vardhan, and G. J. Lal, “A few-shot approach to dysarthric speech intelligibility level classification using transformers,” in *2023 14th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, pp. 1–6, IEEE, 2023.

[6] G. Van Nuffelen, C. Middag, M. De Bodt, and J.-P. Martens, “Speech technology-based assessment of phoneme intelligibility in dysarthria,” *International journal of language & communication disorders*, vol. 44, no. 5, pp. 716–730, 2009.

[7] R. D. Kent, G. Weismer, J. F. Kent, and J. C. Rosenbek, “Toward phonetic intelligibility testing in dysarthria,” *Journal of Speech and Hearing Disorders*, vol. 54, no. 4, pp. 482–499, 1989.

[8] B. Das, K. Daoudi, J. Klempir, and J. Rusz, “Towards disease-specific speech markers for differential diagnosis in parkinsonism,” in *ICASSP 2019-2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 5846–5850, IEEE, 2019.

[9] H. Kim, M. Hasegawa-Johnson, A. Perlman, J. R. Gunderson, T. S. Huang, K. L. Watkin, S. Frame, *et al.*, “Dysarthric speech database for universal access research.,” in *Interspeech*, vol. 2008, pp. 1741–1744, 2008.

[10] J. Kim, N. Kumar, A. Tsiartas, M. Li, S. S. Narayanan, *et al.*, “Automatic intelligibility classification of sentence-level pathological speech,” *Computer speech & language*, vol. 29, no. 1, pp. 132–144, 2015.

[11] A. A. Joshy and R. Rajan, “Automated dysarthria severity classification using deep learning frameworks,” in *2020 28th European Signal Processing Conference (EUSIPCO)*, pp. 116–120, IEEE, 2021.

[12] K. L. Stipancic, K. M. Palmer, H. P. Rowe, Y. Yunusova, J. D. Berry, J. R. Green, *et al.*, ““you say severe, i say mild”: Toward an empirical classification of dysarthria severity,” *Journal of Speech, Language, and Hearing Research*, vol. 64, no. 12, pp. 4718–4735, 2021.

[13] M. S. Paja and T. H. Falk, “Automated dysarthria severity classification for improved objective intelligibility assessment of spastic dysarthric speech,” in *Thirteenth Annual Conference of the International Speech Communication Association*, 2012.

[14] A. A. Joshy and R. Rajan, “Dysarthria severity classification using multi-head attention and multi-task learning,” *Speech Communication*, vol. 147, pp. 1–11, 2023.

[15] S. Gillespie, Y.-Y. Logan, E. Moore, J. Laures-Gore, S. Russell, R. Patel, *et al.*, “Cross-database models for the classification of dysarthria presence.,” in *Interspeech*, pp. 3127–3131, 2017.

[16] E. C. Guerra and D. F. Lovey, “A modern approach to dysarthria classification,” in *Proceedings of the 25th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (IEEE Cat. No. 03CH37439)*, vol. 3, pp. 2257–2260, IEEE, 2003.

[17] A. Hernandez, E. J. Yeo, S. Kim, and M. Chung, “Dysarthria detection and severity assessment using rhythm-based metrics.,” in *INTERSPEECH*, pp. 2897–2901, 2020.

[18] E. J. Yeo, S. Kim, M. Chung, *et al.*, “Automatic severity classification of dysarthria using voice quality, prosody, and pronunciation features,” *Phonetics and Speech Sciences*, vol. 13, no. 2, pp. 57–66, 2021.

[19] K. M. Allison and K. C. Hustad, “Data-driven classification of dysarthria profiles in children with cerebral palsy,” *Journal of Speech, Language, and Hearing Research*, vol. 61, no. 12, pp. 2837–2853, 2018.

[20] S. Gupta, A. T. Patil, M. Purohit, M. Parmar, M. Patel, H. A. Patil, R. C. Guido, *et al.*, “Residual neural network precisely quantifies dysarthria severity-level based on short-duration speech segments,” *Neural Networks*, vol. 139, pp. 105–117, 2021.

[21] C. V. T. Kumar, T. Bhattacharjee, Y. Belur, A. Nalini, R. Yadav, P. K. Ghosh, *et al.*, “Classification of multi-class vowels and fricatives from patients having amyotrophic lateral sclerosis with varied levels of dysarthria severity,” in *Interspeech*, 2023.

[22] F. Rudzicz, A. K. Namasivayam, and T. Wolff, “The torgo database of acoustic and articulatory speech from speakers with dysarthria,” *Language resources and evaluation*, vol. 46, pp. 523–541, 2012.

[23] P. Enderby, “Frenchay dysarthria assessment,” *British Journal of Disorders of Communication*, vol. 15, no. 3, pp. 165–173, 1980.

[24] K. L. Kadi, S. A. Selouani, B. Boudraa, M. Boudraa, *et al.*, “Fully automated speaker identification and intelligibility assessment in dysarthria disease using auditory knowledge,” *Biocybernetics and Biomedical Engineering*, vol. 36, no. 1, pp. 233–247, 2016.

[25] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, *et al.*, “An image is worth 16x16 words: Transformers for image recognition at scale,” *arXiv preprint arXiv:2010.11929*, 2020.

[26] A. Anil, S. V. VV, V. Sowmya, A. Sukumar, M. Krichen, *et al.*, “Influence of spectral bands on satellite image classification using vision transformers,” *Authorea Preprints*, 2023.

[27] C. A. Kumar, A. D. Maharana, S. M. Krishnan, S. S. S. Hanuma, G. J. Lal, and V. Ravi, “Speech emotion recognition using cnn-lstm and vision transformer,” in *International Conference on Innovations in Bio-Inspired Computing and Applications*, pp. 86–97, Springer, 2022.



rized licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:48:43 UTC from IEEE Xplore. Restrictions a